

MAHMUDUL HASAN

Full-Stack Software & Systems Solutions Engineer

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PROFESSIONAL SUMMARY

High-impact **Full-Stack Software & Systems Engineer** with 2.5+ years of production experience architecting concurrent backends (**Java 21 Spring Boot, Rust 2024 Axum**), reactive cross-platform applications (**React 19/TypeScript, Flutter BLoC**), and hardware-level IoT integrations. Proven track record in building enterprise ERP/HRMS suites, smart parking architectures, and native protocol handlers (**ZKTeco ADMS, Frigate AI NVR, RTSP/HTTP CCTV**). Experienced in Linux cloud VPS orchestration, database query optimization (PostgreSQL, SeaORM, Hibernate), and enterprise network/VoIP infrastructure (MikroTik RouterOS, Vicidial PBX). Graduated with top honors (3.75 CGPA) from Dhaka Polytechnic Institute and currently pursuing AMIE (IEB) B.Sc. engineering equivalence.

TECHNICAL SKILLS

Backend & Concurrency:	Java 21 (Spring Boot 3.x/4.x), Rust 2024 (Axum 0.8, Tokio Async Runtime), REST APIs, WebSockets, Spring Security 6, JWT, Argon2
Database & ORM:	PostgreSQL, SeaORM (Async Connection Pools), Spring Data JPA / Hibernate, H2, ACID Transactions, JPA Criteria Specifications
Frontend & Mobile:	React 19 / 18, TypeScript, Flutter (BLoC Architecture), Three.js / React Three Fiber, GSAP, Framer Motion, Tailwind CSS, Base UI, Vite
IoT & Hardware Protocols:	ZKTeco Terminals (ADMS Push Protocol), Frigate NVR (AI Vision REST APIs), CCTV Streaming (RTSP/HTTP), Barcode Multi-Terminal Sync
DevOps & Networking:	Linux Server (Ubuntu), Hostinger VPS, MikroTik RouterOS (Routing, Bandwidth Queues, VLANs), Vicidial VoIP/PBX, Nginx, Docker, cPanel
Libraries & Tools:	Git, GitHub, Apache POI (Excel I/O), OpenPDF, iText7, Metadata-Extractor, Postman, Linux CLI, Bash, Chrome Ext Manifest V3

PROFESSIONAL EXPERIENCE

IT Executive / Software & Systems Engineer — Skylink Innovations Ltd *October 2025 – Present | Dhaka, Bangladesh*

- **Enterprise ERP/HRMS Architecture:** Designed and deployed an all-in-one portal (**Java 21, Spring Boot 3.4, React, PostgreSQL**) for **50+ daily active staff**, featuring RBAC security, dynamic shift schedules, and leave lifecycles.
- **Biometric Attendance Integration (ADMS):** Developed native AdmsService to handle real-time HTTP push communication from ZKTeco biometric terminals, eliminating manual Excel logs and human error.
- **Automated Payroll Engine:** Implemented a zero-error payroll module with multi-parameter JPA Specifications, automating salary disbursements, deductions, and dynamic PDF/Excel report generation via **OpenPDF** and **Apache POI**.
- **Operational Micro-Tools & Telemetry:** Built a **Photo EXIF metadata analyzer** (extracting GPS and device timestamps), an interactive **Crew Coverage Map** (ZipCodeGeoService), and real-time monitoring via WebSockets and Web Push.
- **Web Platform Deployments:** Engineered and deployed **7+ corporate web platforms** using **React 19, Three.js**, and **Tailwind CSS** on cloud VPS instances with Nginx reverse proxy and SSL configs.
- **Corporate IT & VoIP PBX:** Administered MikroTik core routers, managed switches, corporate email infrastructure, and a high-availability **Vicidial BPO telephony cluster**.

Software Executive / IoT & Systems Engineer — SPY Security Solutions*March 2024 – October 2025 | Dhaka, Bangladesh*

- **Smart Parking Management System (PMS-V2):** Architected a multi-terminal commercial parking automation platform with high-concurrency **Rust (Axum 0.8 + Tokio)** backend and **Flutter (BLoC)** cross-platform frontend.
- **Hardware & Terminal Integration:** Integrated ZKTeco gate barrier controllers via **ADMS**, embedded live CCTV feeds over **RTSP/HTTP**, and engineered barcode scanner validation for multi-terminal checkouts.
- **Async Tariff & Anti-Passback Engine:** Built dynamic second-by-second tariff computation, anti-passback rule enforcement, and duplicate-scan mitigation using **SeaORM** and PostgreSQL connection pools.
- **AI Computer Vision (Frigate NVR):** Deployed Frigate NVR with computer vision object detection and REST API webhooks for automated surveillance, vehicle detection, and VMS troubleshooting.
- **Edge Network Infrastructure:** Designed physical PoE network topologies and subnet routing across cashier terminals and entry/exit gates to ensure fault-tolerant hardware connectivity.

KEY ENGINEERING PROJECTS & LIVE DEPLOYMENTS

PMS-V2 (Smart Parking Management System) | Rust (Axum, Tokio), SeaORM, Flutter, PostgreSQL, Docker [GitHub]

- High-concurrency parking automation platform featuring dynamic tariff calculations, multi-terminal cashier synchronization, automated barrier control, and real-time RTSP video monitoring.

Enterprise Skylink HRMS & Operations Suite | Java 21, Spring Boot 3.4, React, PostgreSQL, WebSockets [GitHub]

- Enterprise platform with ZKTeco biometric sync, automated payroll computation, GPS photo EXIF extraction, and push notifications.

Live Commercial Web Deployments | React 19, TypeScript, Three.js, GSAP, Framer Motion, Tailwind CSS

- **Skylink Official Portal:** skylinkltd.ai (Interactive motion scenes with GSAP & Framer Motion)
- **Finara Property Solutions LLC:** finaraprosolutions.com (React 19 & Three.js 3D rendering)
- **Homesync LLC:** homesyncllc.org (Interactive 3D Canvas with React Three Fiber)
- **Dream Zone Builders Inc:** dreamzonebuilders.com (Base UI & Tailwind CSS)

Watch Employee & Activity Monitor | Java 21, Spring Boot, Chrome Extension (Manifest V3), OpenAPI [GitHub]

- Telemetry monitoring solution tracking workstation active tabs, idle states, and credential autofill via background service workers.

EDUCATION & PROFESSIONAL QUALIFICATIONS

B.Sc. in Engineering (Equivalence) — Section A

The Institution of Engineers, Bangladesh (IEB) — AMIE | Enrolled 2026 (In Progress)

Diploma in Engineering (Computer Science & Technology)

Dhaka Polytechnic Institute (DPI) | Session: 2021–2022 | CGPA: 3.75 / 4.00 (Distinction)

PERSONAL INFORMATION & REFERENCES

Father's Name:	Robiyal Hossen	Engr. Niger Sultana
Mother's Name:	Lutfur Nahar Begum	Head of Department (HOD), CST Dept
Date of Birth:	15 February, 2002	Dhaka Polytechnic Institute (DPI)
Gender / Status:	Male Married	☎ +880 1812-112664
Blood Group:	B+ (Positive)	Nazrul Hasan Raju
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